

Setup Guide - Telegram Voice-to-Text n8n Workflow

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A complete, step-by-step guide to setting up ngrok, connecting your Telegram bot, and configuring the Groq API inside n8n.

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1. Installing & Configuring ngrok

ngrok creates a secure, public HTTPS tunnel to your local machine. Telegram requires an HTTPS webhook URL to deliver messages, so this step is mandatory when running n8n on localhost.

Step 1.1 — Create an account Go to <https://ngrok.com> and sign up for a free account.

Step 1.2 — Download ngrok From the ngrok dashboard, download the version for your operating system (Windows/Mac/Linux) and unzip it to a folder you can easily find, e.g. `C:\ngrok\`.

Step 1.3 — Get your authtoken On the ngrok dashboard, go to **Getting Started → Your Authtoken**. Copy the token shown there.

Step 1.4 — Add the authtoken to your local ngrok installation Open a terminal in the folder where you extracted ngrok, and run:

```
ngrok config add-authtoken YOUR_AUTH_TOKEN_HERE
```

You should see a confirmation message that the configuration file was updated. This step only needs to be done once.

2. Establishing the Tunnel

Step 2.1 — Start n8n first (in its own terminal) Before starting the tunnel, make sure n8n is running normally on its default port:

```
n8n start
```

Confirm it's accessible at `http://localhost:5678` in your browser.

Step 2.2 — Open a second terminal window Keep the n8n terminal running, and open a **new, separate terminal**

window for ngrok.

Step 2.3 — Start the tunnel In the new terminal, run:

```
ngrok http 5678
```

Step 2.4 — Copy your public URL ngrok will display a screen similar to this:

```
Session Status      online
Account             your-email@example.com
Version             3.x.x
Region              India (in)
Forwarding           https://a1b2-3c4d-5e6f.ngrok-free.app -> http://localhost:5678
```

Copy the full `https://...ngrok-free.app` URL — you'll need it in the next step.

Important: On the free ngrok plan, this URL changes every time you restart the tunnel. You'll need to repeat Steps 2.3-2.4 and update your `WEBHOOK_URL` (Step 3.2) each time you restart your machine or ngrok.

Step 2.5 — Keep both terminals open Do not close either the n8n terminal or the ngrok terminal while testing — both must stay running simultaneously.

3. Connecting n8n to the Tunnel

Step 3.1 — Stop the currently running n8n instance Go back to your n8n terminal and stop it with `Ctrl + C`.

Step 3.2 — Restart n8n with the webhook URL and binary mode set

Using **Command Prompt**:

```
set WEBHOOK_URL=https://a1b2-3c4d-5e6f.ngrok-free.app/
set N8N_DEFAULT_BINARY_DATA_MODE=memory
n8n start
```

Using **PowerShell**:

```
$env:WEBHOOK_URL="https://a1b2-3c4d-5e6f.ngrok-free.app/"
$env:N8N_DEFAULT_BINARY_DATA_MODE="memory"
n8n start
```

Replace the URL with **your own** ngrok forwarding address from Step 2.4. Keep the trailing `/` at the end.

Step 3.3 — Verify the webhook is registered correctly Open `http://localhost:5678` in your browser, open your workflow, click on the **Telegram Trigger** node, and click **Test this trigger**. If everything is configured correctly, it will show “Waiting for trigger event” without any “Bad Request” errors.

4. Creating Your Telegram Bot

Step 4.1 — Open Telegram and search for BotFather In the Telegram app (mobile or desktop), search for the official account **@BotFather** and open a chat with it.

Step 4.2 — Create a new bot Send the command:

```
/newbot
```

Step 4.3 — Name your bot BotFather will ask for: - A **display name** (e.g. “My Voice Transcriber”) — this can be anything - A **username** ending in `bot` (e.g. `my_voice_transcriber_bot`) — this must be unique across all of Telegram

Step 4.4 — Copy your bot token Once created, BotFather will reply with a message containing your **bot token**, which looks like:

```
123456789:AAExampleTokenStringHere12345678
```

Copy this token and store it somewhere safe — this is what n8n will use to control your bot.

Step 4.5 — Start a chat with your bot Search for your bot’s username in Telegram and send it any message (e.g. “Hi”). This is required — n8n cannot receive messages from a bot that has zero chat history.

5. Connecting Telegram Credentials in n8n

Step 5.1 — Open the Telegram Trigger node In your n8n workflow, double-click the **Telegram Trigger** node.

Step 5.2 — Create new credentials Click the **Credential** dropdown → **Create New Credential**.

Step 5.3 — Paste your bot token In the credential form, paste the bot token you copied from BotFather into the **Access Token** field.

Step 5.4 — Save and test Click **Save**. Back on the node, set **Trigger On** to `Message`, then click **Test this trigger**. Send a voice note to your bot in Telegram — it should appear in the output panel within a few seconds.

6. Getting a Groq API Key

Step 6.1 — Create a Groq account Go to <https://console.groq.com> and sign up (free).

Step 6.2 — Navigate to API Keys On the left sidebar, click **API Keys**.

Step 6.3 — Create a new key Click **Create API Key**, give it a name (e.g. “n8n-voice-transcription”), and click **Submit**.

Step 6.4 — Copy the key immediately Groq only shows the full key once. Copy it and store it somewhere safe — you won’t be able to view it again later (though you can always generate a new one).

Step 6.5 — Confirm Whisper model access Go to **Settings → Limits** (or your project settings) and confirm `whisper-large-v3-turbo` appears as an allowed model. If it’s greyed out, request access or use the “Select Models” screen to enable it for your project.

7. Connecting Groq Credentials in n8n

Step 7.1 — Open the HTTP Request node Double-click the HTTP Request node used for transcription.

Step 7.2 — Set the Authentication type Set **Authentication** to `Generic Credential Type`, then set **Generic Auth Type** to `Header Auth`.

Step 7.3 — Create the credential Click **Set up credential** (or the dropdown → **Create New Credential**).

Step 7.4 — Fill in the header details - **Name:** `Authorization` - **Value:** `Bearer YOUR_GROQ_API_KEY`

Replace `YOUR_GROQ_API_KEY` with the actual key from Step 6.4, keeping the word `Bearer` and the space before your key.

Step 7.5 — Save the credential Click **Save**. This credential will now be reusable across any other HTTP Request nodes that need Groq access.

Step 7.6 — Verify the connection Click **Execute step** on the HTTP Request node (with a voice file already flowing into it from earlier nodes). A successful response returns a JSON object containing a `text` field with your transcribed speech.

8. Final Verification Checklist

Use this checklist to confirm every part of the setup is correctly connected before considering the workflow production-ready.

#	Check	Status
1	ngrok authtoken configured	☐
2	ngrok tunnel running and forwarding to port 5678	☐
3	n8n restarted with <code>WEBHOOK_URL</code> and <code>N8N_DEFAULT_BINARY_DATA_MODE</code> set	☐
4	Telegram bot created via BotFather, token copied	☐
5	Telegram credential saved in n8n	☐
6	Telegram Trigger successfully receives a test voice note	☐
7	Groq account created, API key generated	☐
8	Whisper model enabled for the Groq project	☐
9	Header Auth credential saved in n8n with correct Bearer token	☐
10	HTTP Request node returns a transcript successfully	☐
11	Full workflow activated (toggle set to "Active")	☐
12	End-to-end test: voice note sent → transcript received back in Telegram	☐

Once every box above is checked, your Telegram Voice-to-Text pipeline is fully operational and ready for daily use or further extension.